

Privacy for All: Empowering Vulnerable Groups with Diversity-Oriented Online Protection

Clara Strathmann
University of Duisburg-Essen
Duisburg, Germany
clara.strathmann@uni-due.de

Abstract

Vertical data flows (e.g., those directed toward companies) are challenging to understand even for technically knowledgeable adults, exposing them to privacy risks. Vulnerable groups with limited experience or cognitive abilities face even greater risks. To support these users in protecting their privacy online, it is essential to understand their perceptions of data flows and privacy needs. This dissertation employs a mixed-method participatory approach involving children, seniors, and individuals with cognitive disabilities. Initial focus groups have explored their understanding of vertical data flows, privacy risks, and protective measures. Building on these insights, co-creation workshops aim to develop tangible cues that enhance privacy awareness, literacy, and self-efficacy with minimum cognitive load. The effectiveness of these tools will be tested longitudinally, allowing iterative improvements to better suit the needs of these vulnerable groups. The goal is to achieve comprehensive and diverse empowerment, ensuring tailored privacy protection for everyone.

CCS Concepts

• **Security and privacy** → Human and societal aspects of security and privacy; Social aspects of security and privacy; • **Social and professional topics** → User characteristics; People with disabilities; User characteristics; Age; • **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

Keywords

vulnerable groups, vertical privacy, participatory approach, privacy-enhancing technologies

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1 Research description

1.1 Background and Motivation

The digital environment poses significant challenges to privacy, as individuals constantly generate data during everyday activities, from ordering food to engaging on social media platforms. Data shared online or inferred from digital traces provides companies with valuable insights that can be used to personalize user experiences, increasing engagement but often compromising user privacy [25]. While some data is knowingly disclosed, much of the information is collected without users being fully aware of the implications, particularly when it comes to vertical or institutional data flows – data captured by organizations and used in ways that are often opaque to the individual [11].

Vertical privacy, which refers to the flow of information from individuals to institutions such as companies, government bodies, or other organizations, presents unique challenges that go beyond the realm of horizontal privacy, where data is actively shared with peers. Unlike horizontal privacy, which involves conscious self-disclosure on social platforms (e.g., posting updates or sharing photos), vertical privacy often entails passive data extraction through means such as tracking cookies, metadata, and browsing history analysis. This form of data collection is largely invisible to users and gives institutions deeper insights into their behavior, with limited awareness or control on the part of the individual. Consequently, managing vertical privacy is more difficult for users, as it requires resources such as privacy literacy skills to navigate complex settings and understand abstract data flows, alongside the time and ability to grasp the implications of terms and conditions [5].

Certain groups are especially vulnerable to privacy risks due to relevant online experiences or cognitive limitations that hinder their ability to understand and manage vertical data flows effectively. For instance, primary school children, seniors, and individuals with cognitive disabilities (CD) face significant obstacles that affect their privacy literacy, making it more difficult for them to turn theoretical knowledge into protective behaviors [10].

For children, digital devices and online media have become integral, with some as young as eight having social media profiles and engaging with online platforms [8, 14]. Despite being active users, children's understanding of data flows is often limited to what is visually present or concrete, struggling to grasp abstract concepts like vertical data flows [9]. Research has shown that children hold misconceptions about data collection, believing it is only for enhancing user experiences or that data remains with a single platform [26]. Furthermore, while they might express positive attitudes toward digital surveillance, these views might shift as their knowledge of data practices increases through targeted interventions [17, 21]. This highlights the role of knowledge in shaping children's privacy

concerns, which are typically less pronounced when they do not perceive a direct personal impact [26].

Seniors, typically defined as individuals over the age of 60, also face distinct privacy challenges due to age-related cognitive declines, slower technology adoption rates, and a lack of familiarity with rapidly changing digital environments [18, 19]. Although many older adults use the internet for everyday activities such as communication and information-seeking [20], their understanding of privacy concepts such as “cookies” or “privacy policies” is often limited [15]. Seniors tend to hold negative attitudes towards targeted ads, mainly due to their prevalence, yet some find them useful, indicating mixed perceptions of vertical data flows [28]. Additionally, while older adults express higher privacy concerns compared to younger users, they are less likely to engage in protective behaviors, underscoring the need for tailored approaches that go beyond privacy literacy education alone [24, 27].

Individuals with CD, including those with intellectual and learning disabilities, face unique challenges in navigating digital privacy. These disabilities can impact memory, attention, language comprehension, and the ability to understand abstract concepts, making it difficult for people with CD to grasp the implications of vertical data flows [3, 23]. Research suggests that people with CD generally have limited knowledge about online data practices, which may contribute to a higher willingness to share personal information without considering potential privacy threats [4]. Their conceptualization of privacy often revolves around tangible experiences rather than abstract data processing, with some studies noting that individuals with CD tend to be more open to sharing information in real-life contexts, particularly for health-related reasons [2].

The privacy risks associated with vertical data flows are thus not uniformly experienced, with vulnerable populations like children, seniors, and people with CD facing distinct challenges that traditional privacy literacy efforts may not adequately address. The expectation that all users can be sufficiently informed to make protective decisions is increasingly questioned, as privacy literacy places the burden of protection primarily on individuals, neglecting structural and institutional factors that shape privacy behaviors and risks [6]. As Meier and Bol [13] suggest, merely being aware of the distinction between horizontal and vertical privacy can influence how people perceive privacy risks, which can, in turn, affect their protective behaviors [22]. Despite its significance, research has shown that users generally do not have a pronounced understanding of how data flows on the internet. Therefore, new approaches are needed to protect vulnerable groups who are less likely to benefit from traditional privacy literacy efforts due to cognitive, situational, and socio-economic barriers.

1.2 Specific Research Objectives and Differentiation From Related Work

The primary aim of this research is to develop a comprehensive framework for inclusive privacy protection that addresses the limitations of current privacy mechanisms and extends support to vulnerable populations. The following specific objectives guide this research:

- Assessing privacy knowledge and attitudes across vulnerable groups: The study seeks to understand how different

vulnerable groups – children, seniors, and adults with CD – perceive and manage privacy in the context of vertical data flows.

- Exploring alternatives to current cognition-focused approaches: We aim to evaluate whether sensory and contextual privacy signals can reduce the cognitive load associated with understanding complex privacy policies and other privacy-protecting mechanisms, thereby supporting more intuitive privacy protection.
- Developing and testing inclusive privacy interventions: Using a mixed-methods approach, we seek to design, test, and refine privacy interventions tailored to the unique needs of diverse user groups.

The landscape of privacy research is rich with studies that have addressed various aspects of privacy protection, yet our study distinguishes itself in significant ways. Firstly, traditional privacy literacy efforts have aimed to educate users about privacy risks and protective behaviors, yet they often fall short for vulnerable populations who face unique challenges. Privacy literacy assumes a level of cognitive ability and access to resources that not all users possess. Our work goes beyond merely informing users about privacy; we investigate how external design interventions, such as sensory cues and contextual signals, can help evoke protective behaviors without requiring extensive cognitive deliberation. This approach aligns with Acquisti et al.’s [1] argument that privacy protection should not solely rely on cognitive processes but should involve visceral, real-world experiences that evoke immediate responses to privacy threats.

Secondly, our study explicitly integrates diversity considerations into the development of privacy protection mechanisms. This includes the systematic incorporation of diverse vulnerabilities, such as CD into privacy strategies. While several studies have examined privacy attitudes among general populations, there is limited research on how vulnerable groups, such as children, seniors, and adults with CD, perceive vertical data flows and manage their privacy. Our research fills this gap by conducting a comparative analysis of these groups’ privacy knowledge and attitudes, identifying areas where tailored support and interventions are needed. By understanding the differences in privacy perspectives across groups, we aim to develop more inclusive and adaptive privacy protection measures, eventually suited for everyone.

1.3 Research Approach

These insights are enabled by employing a mixed-methods approach that integrates quantitative and qualitative data collection and analysis. The study consists of the following phases: Firstly, nine focus groups were conducted to gather qualitative insights from vulnerable populations regarding their knowledge about various privacy-related aspects, focusing on vertical data flows. This will be followed by participatory design workshops to co-develop tailored privacy protection measures, which will then undergo iterative testing and refinement to enhance privacy behaviors and reduce privacy-compromising actions.

1.3.1 Focus Groups. The focus group study explored vulnerable populations’ knowledge about and attitudes toward vertical data

flows as well as risks and protection measures. Investigated vulnerable groups included children, seniors, and adults with CD, alongside two control groups of adults without CD. Conducted between April and June 2024, nine focus groups were employed as the primary qualitative research method, given their suitability for encouraging discussion among participants with similar demographic backgrounds and for addressing exploratory research questions [16]. The study was preregistered, and approval was obtained from the local ethics committee.

Participants were recruited through various channels, such as relevant institutions (e.g., primary schools, senior centers, associations supporting independent living) and community outreach (e.g., newspaper calls, social media posts). The inclusion criterion for all groups was the regular use of internet-based applications. Focus group sizes of three to six participants were chosen, as recommended in previous studies involving vulnerable groups [7]. The nine focus groups included two groups of primary school children, two of seniors aged 65 and above, three of adults with CD (with varying levels of independence), and two control groups.

A semi-structured interview guide facilitated the discussions. Topics included participants' knowledge about data collection and processing, perceived benefits and risks of data sharing, and online risks and protection measures, such as cookie banners. The example of YouTube was used to ensure the discussions were relevant to all participants, and a YouTube video of the participants' choice served as a visualization tool. Data were collected through audio recordings, transcribed, and analyzed using MaxQDA 2024 for qualitative content analysis, adhering to Mayring [12]. Initially, two transcripts were analyzed independently to identify relevant themes. An iterative process among four researchers led to the development of a final coding scheme, which was applied to all transcripts. Intercooder reliability scores ranged from 0.58 to 0.88, indicating good consistency across the coding process.

Informed consent was obtained from all participants, with guardians providing consent for children and some adults with CD as required. Participants received a 50 € compensation, which varied by group: toy store vouchers for children, grocery vouchers for adults with CD, and cash for the seniors and control groups.

1.3.2 Co-Creation Workshops and Iterative Testing. An iterative design process builds on the findings of the focus group study. Specific vulnerable groups' needs that have been identified in the focus groups will be the foundation of developing cues for supporting privacy management through participatory efforts. Traditional concepts like privacy literacy often fall short in addressing these needs. The overarching goal is to develop a measure that can enhance media literacy, privacy behavior, and users' awareness of privacy risks in the digital landscape as well as sense of self-efficacy in managing their privacy. These measures will then be evaluated through a longitudinal study with several measurement points to assess their impact on privacy behaviors and improve them accordingly. This approach aims to achieve several key objectives, including the participatory creation of helpful cues for privacy management and testing their effectiveness of these cues in increasing privacy awareness among vulnerable groups.

1.4 Results and Contributions to Date

We observed key similarities in how children and adults with CD understand vertical data flows within the focus group study. Both vulnerable groups exhibited a general lack of knowledge, with their understanding largely based on observable and tangible processes. In terms of attitudes, children did not demonstrate strong opinions toward datafication resulting from vertical data flows, potentially due to limited understanding. Adults with CD, however, showed attitudes that appeared to be influenced by their caregivers. In contrast, seniors exhibited a more cynical perspective, but they heavily relied on their life experience, which could lead them to underestimate the capabilities of big data and the potential effects of personalized ads.

These findings have influential implications. To make data processes more understandable, it is important to use visual and tangible cues that help convey abstract privacy concepts, such as the one we aim to develop in the upcoming co-creation workshops. Tailored educational approaches should address the specific needs of different groups: for children, using interactive tools can facilitate learning; for seniors, fostering critical thinking skills is essential; and for adults with CD, developing reflective skills can enhance their understanding of data privacy. Additionally, educating caregivers and reference persons is crucial, as they play a significant role in shaping the attitudes and knowledge of vulnerable individuals. Mass media also has a responsibility to represent data security in a realistic manner, avoiding the spread of misconceptions. Since media significantly influences users' mental models, accurate portrayals can help shape a more informed public perception of data privacy issues.

By means of this focus group study, we could compare conceptions and attitudes of different vulnerable groups toward vertical data flows. With this, we aimed to encourage future research to explore synergies across studies involving populations with similar vulnerabilities, such as children and adults with CD. While targeted approaches are essential, there is value in sharing insights across groups that are typically treated separately. Furthermore, individuals with CD are significantly understudied, particularly in the context of privacy online. Our study provided a systematic, in-depth evaluation of adults with CD, examining both their mental models and attitudes toward vertical data flows. We identified significant knowledge gaps and variations based on the degree of CD, which can inform strategies to empower individuals with CD to navigate the internet more safely.

1.5 Expected Next Steps and Open Questions

Next steps include a qualitative content analysis of the focus groups' second part, namely the vulnerable groups' knowledge of risks and protection measures within privacy online, and potential ideas and wishes for improvement of such measures. Together with our findings on the groups' knowledge about vertical data flows, this will provide the foundation for engaging with them in the upcoming co-creation workshops. Within these, the goal is to collaboratively design cues and prompts aimed at enhancing privacy awareness, media literacy, and privacy-related behaviors. These cues will be evaluated and refined based on user feedback and their effectiveness in increasing privacy awareness. A longitudinal study

will investigate the effectiveness of these cues over time. Open questions include determining the most effective types of cues for different user groups, understanding how context influences the perception and impact of these cues, and identifying the best methods for measuring changes in users' privacy behaviors and self-efficacy. Further research is also needed to explore how the timing and delivery of cues affect users' ability to manage privacy risks proactively.

The main challenge in designing the co-creation workshops is to ensure that vulnerable groups are empowered to contribute their ideas and perspectives effectively, allowing them to develop tools and cues that support their privacy protection. To achieve this, the workshops need to create a safe, inclusive, and empathetic environment where participants feel comfortable sharing their experiences and needs. It is crucial to prepare application scenarios that reflect realistic and relevant privacy risks for each group, using materials and approaches tailored to their specific abilities and knowledge levels. Interactive formats such as role-playing, storytelling, or visual aids could facilitate creative exchanges and help participants develop concrete measures for safeguarding their privacy. Additionally, continuous feedback should be gathered to adapt the workshop process to the evolving needs of the groups, ensuring that their preferences and ideas are integrated as effectively as possible.

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